



Malware costs COs \$491Bn

New study says malware related issues due to pirated software could cost companies \$491 Billion dollars <http://dgit.in/MalwareMoney>



NASA to capture an asteroid

The firm has unveiled a plan to capture an asteroid and then bring it back to earth for research <http://dgit.in/NASARock>

Tomorrow's tech

ligence which is new thoughts, new ideas, creativity, etc. These are things that we don't even understand fully among us humans yet, much less how to get that into a robot.

1 Do you think there's a lack of participation when it comes to academic research concerning the field of robotics? Not enough students interested in robotics?

JM: When it comes to academic research, I see robotics as a good way to attract students to computer sciences, mechanical engineering, electrical engineering, or technology in general. There aren't as many people as I'd have liked, but the number of students taking robotics is increasing. What excites me the most right now is the Maker Movement, where there's this huge DIY resurgence, which is helping the robotics cause. When I started teaching, I'd get students who didn't know how to use a screwdriver, but now I work with students who tell me that the chips we're working with in the labs are out of date. This is very encouraging from the academics perspective, to see that cultural shift that's making it cool to do technology. Technical interest in the community at large is also going up which is an encouraging trend. It's a very exciting time in the field of robotics right now.

1 What's your impression of Indian students?

JM: I'm biased because the students I meet are always the top students from India. I'm always impressed with the Indian students I get to interact with. My perception of the culture in Asia, India in particular, is that education's respected here. There's much more desire to learn, and the social costs of being interested in technology are lower, and the possibility of being teased and ridiculed in the locker room for playing with robots or video games or studying maths is still a problem in the States. The choices faced by kids when they reach middle-school at the age of 15 or 16 years, deciding whether to do something they enjoy (which is building mechanical robots, studying software, or electronics) or do something to look cool. Cool wins more than it ought to right now. We want to tell students about Geek Chic – that you can do both. You can

be cool, you can do technology, and the two things don't need to be mutually exclusive.

1 The digital age has evolved to include previously off-the-grid machines. How do you see the field of robotics or AI adjusting to this shifting landscape?

JM: The pervasiveness of technology, with smart machines, and the internet of things, is definitely advancing rapidly. The simultaneous growth of machine learning (which is different from AI) and predictive analysis of behaviour where the computer can look at what you've been doing and take a good guess of what you'll be doing next is already a reality -- in some cases like Amazon's and online music streaming services' recommendation engines, for e.g. We will see this behaviour increasingly and we might not even notice it. The phones will get smarter, cars will get smarter, your refrigerator will send you an email when you're running short of milk,

that are going to take over the world, you can't do that if you trip over your shoelaces.

1 How has sci-fi influenced the field of robotics and AI?

JM: So there's sci-fi and there's science reality, and they both advise each other at some level. Where science fiction inspires people, wakes them up when they're 10-11 and they get old and they realize how hard it is. It's still fun, but you realize how daunting the technical challenges are. As an inspiration, a well-informed sci-fi can be really very impressive in terms of taking things forward 50-60 or a 100 years forward. This is where Asimov is a true inspiration. In 2014, his writings feel a bit dated because of the way they're talking and doing things, but even after 60 years his logic, his principles make sense... man, he nailed so many things.

So yes, sci-fi inspires us, and the truly good sci-fi leads us to want to strive for



"We want to tell students about Geek Chic – that you can do both. You can be cool, you can do technology, and the two things don't need to be mutually exclusive."

and these things will happen bit by bit. The first smart refrigerator will be heralded as a breakthrough, as RFID tagged food becomes the norm in our house, and there will be a few media stories around it, and it will fade. We'll just get used to it.

1 But is there an inherent danger of things getting out of hands and robots taking over the world?

JM: There are two reasons I'm not worried about that possibility. The first reason is a philosophical argument: If we create this thing, a sentient, self-aware robot, why would it want to exterminate its creators? That's not logical, with an assumption that we create something that's compassionate and caring, something that isn't sadistic in nature. The second reason why I'm not worried about the possibility is because there are currently no robots that can tie shoe laces, there are no robots who can safely cross streets. If we're going to have robots

more. Science reality is what we have in the present, the slow careful work doing basic research, figuring out good applications, and applying that to building products that we can actually use. And each of these phases have separate challenges. The basic research is what I'm focusing now as a professor, but with an eye towards products that people can use, particularly towards education. Advising the research about the systems that have been doing it for 160 million years -- ants and bees are about that old. The algorithmic solutions that nature's devised to these things are critical to our understanding. We know ants and bees are very successful in how they forage, reproduce, and get around the world. We just need to figure out a way to translate all of that into robots.

For Mr. McLurkin's biography, visit <http://dgit.in/ljqukQY>. Also, don't miss to catch the detailed version of this interview at: <http://dgit.in/JamesMcLurkin>